Henry Glover

9/16/2024

MASC-310 Monday Lab

Analyzing Work Hardening Data – Indent #1

Table-

Indent number analyzed: (1)

Test force (gf): 300 **Dwell Time (sec):** 10

Condition	%R	ε_T	d_1 (µm)	d_2 (µm)	d_{ave} (µm)	HV	H (MPa)	Estimated σ_y
As received	0	0	69.0	70.0	69.5	115.17	1129.15	376.38
2 turns	2.778	0.028	63.0	62.0	62.5	142.42	1396.69	465.56
4 turns	13.89	0.149	56.0	56.0	56	177.39	1739.74	579.91
6 turns	23.45	0.267	54.4	57.2	55.8	178.67	1752.24	584.08

Plot-

Multipart Figure-

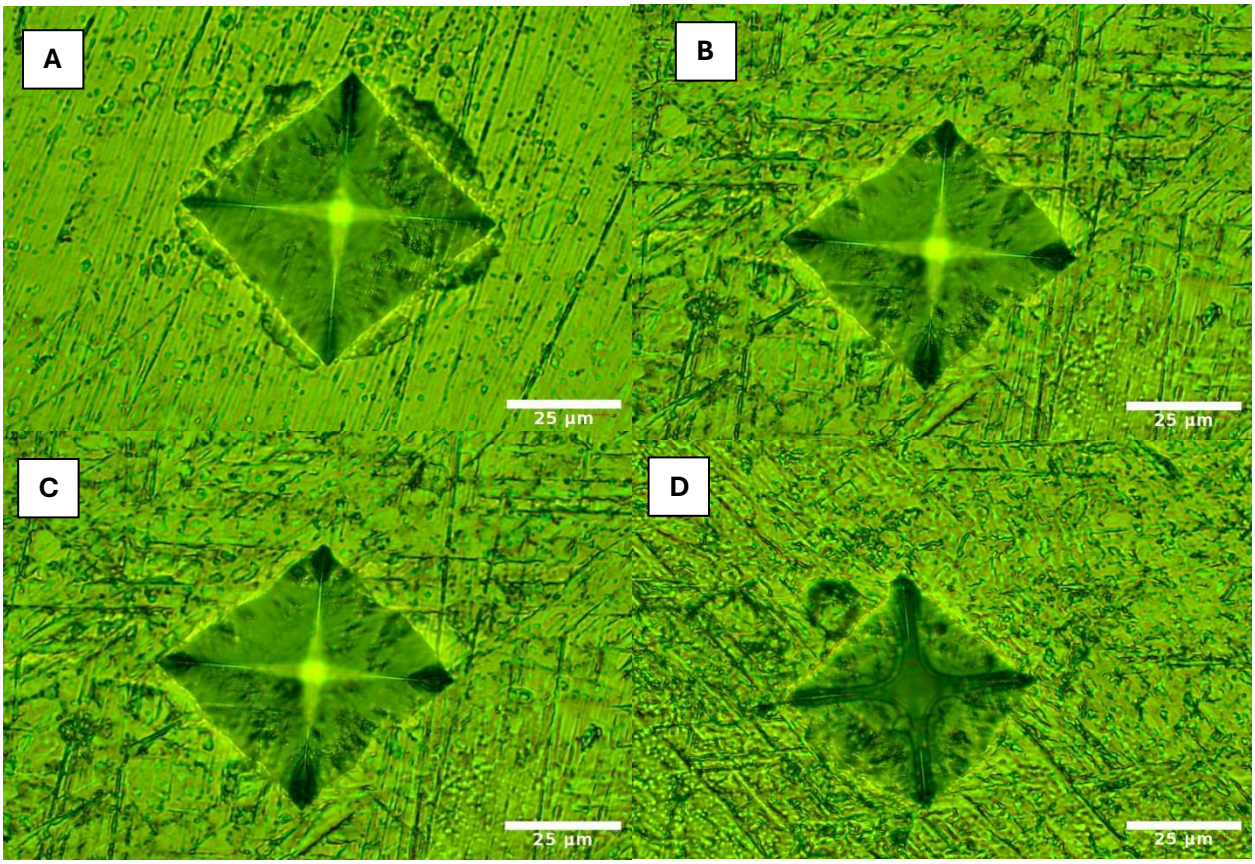

Figure 1 – Representative indents for brass in the (A) as received condition and after (B) 2 turns, (C) 4 turns, and (D) 6 turns. The scale for all images is equivalent because the images were taken at the same magnification level.